

Drag to Tune

We unlocked editing inside the analysis modes. The graphs now change under your hand — which only means anything if you can still see what they were.

Status shipped to staging Asking for feedback on the visuals, not the behaviour

Surfaces graph panel · canvas · transport

00 What changed

PMKS+ is a planar linkage simulator used for teaching. Its two analysis modes — Kinematic and Force — used to be read-only: you built a mechanism in Edit, switched over, and looked at graphs of position, velocity, acceleration and reaction force across one cycle of the machine.

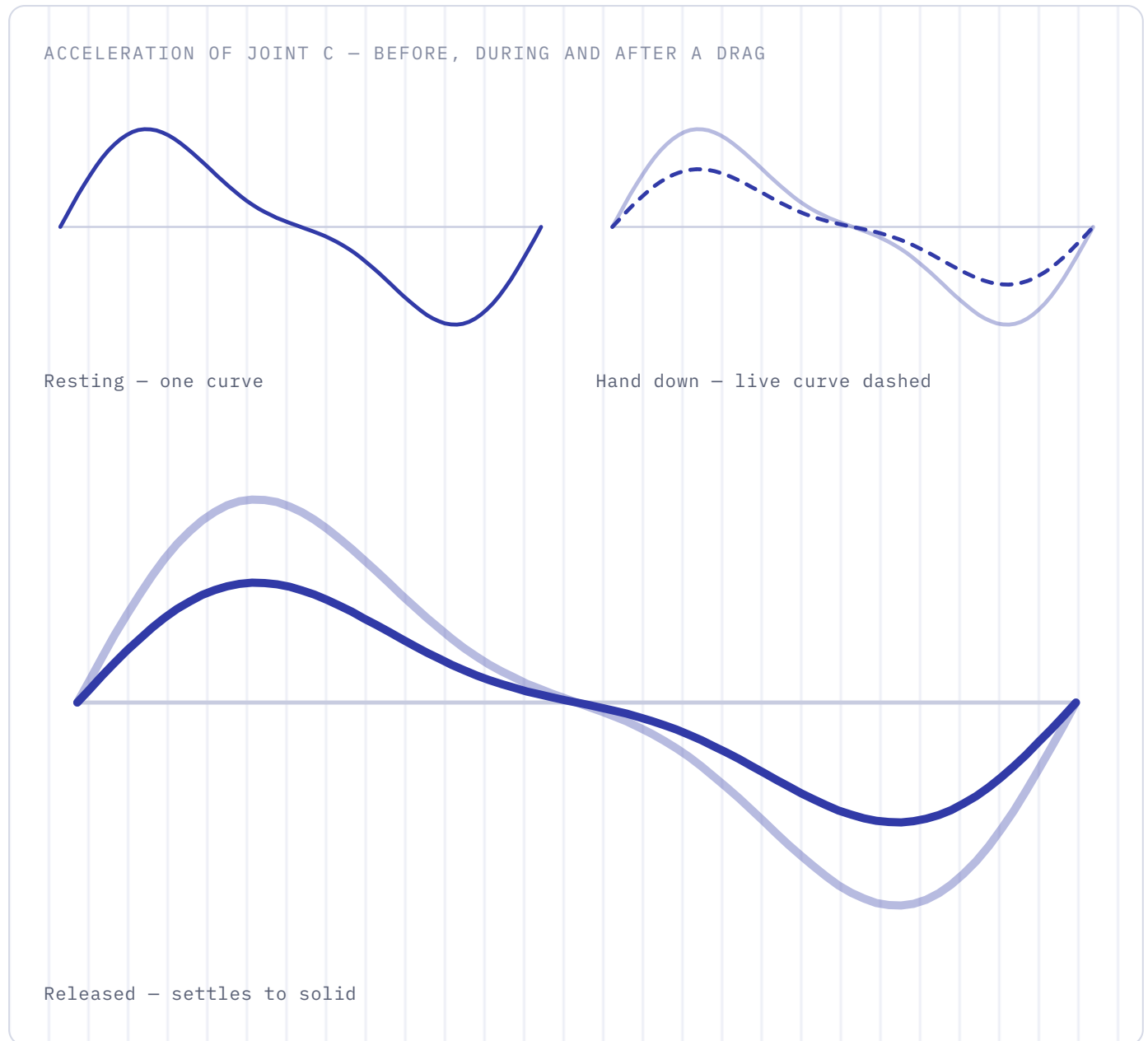
That lock is gone. You can drag a joint while the graphs are open and they re-solve as you move. The workflow it exists for is one sentence:

Watch the acceleration peak, move the coupler pivot, watch the peak come down.

So the design problem is **comparison**, and the numbers that come out of it.

01 The comparison overlay

When a drag begins, the graph snapshots every curve currently plotted and keeps drawing it underneath the live one.



	BEFORE CURVE	LIVE CURVE
Colour	the series' own colour at 35% alpha	the series' own colour, full
Stroke	2px solid	2px — dashed (5) while the hand is down, solid on release
Draw order	behind	in front
Legend	X before, Y before, Z before	X, Y, Z

Two encodings are doing work, deliberately. **Ghosting means “earlier”** — the same word the canvas uses for the start-pose ghost, so a reader learns it once. **Dashing means “provisional”**, and applies only while your finger is down; on release the curve becomes an ordinary solid line, because it is no longer a preview. It is what the mechanism does now.

Lifecycle

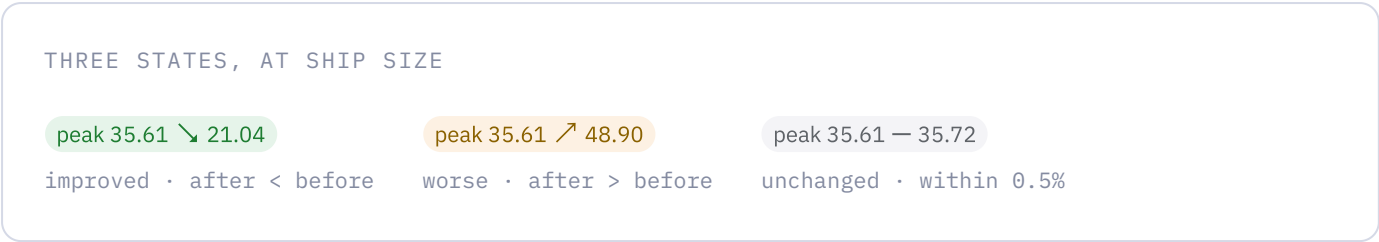
One baseline per gesture. It survives the release — the tuning loop is drag, look, drag again, and taking the comparison away at the moment you finally have a still frame to read would serve the code, not the reader. It is replaced by the next drag, cleared by the **Clear** button, and dropped on undo or redo.

The y-axis rule

The axis freezes to the range the *before* curve was drawn to, and **widens but never shrinks** while you drag. Recomputed per frame it swam under the very curve you were watching — a drag through a toggle sent it to 30,000 and the next frame brought it back to 800, and every frame looked identically tall. Outlier trimming (1st–99th percentile) applies only to the live curve, and only when the full spread exceeds 4× the inner spread: acceleration near a toggle position is a true singularity, and two samples out of 360 reading twenty thousand will flatten a real 0–12 curve onto the zero line.

02 The peak readout

Two curves is squinting. Two numbers and an arrow is reading. So the number the whole overlay exists to move gets said out loud, above the plot.



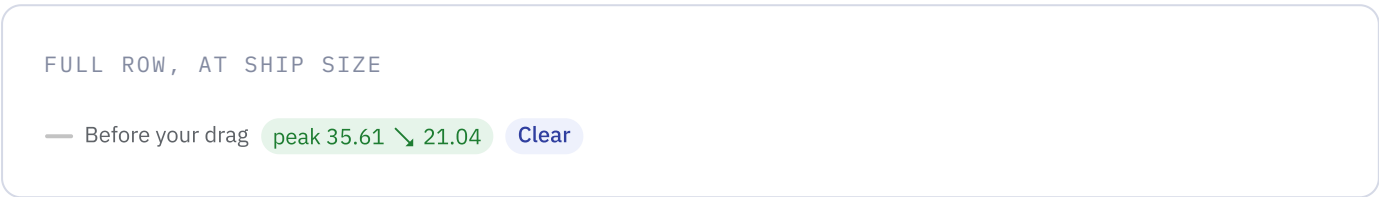
STATE	RULE	BACKGROUND	TEXT	GLYPH
Improved	after < before	#e6f4ea	#1e7d32	south_east
Worse	after > before	#fdf1e3	#8a6100	north_east
Unchanged	within 0.5%	#f4f4f7	#5f6368	drag_handle

The 0.5% deadband exists because a rounding difference painted green is a lie — an arrow claiming an improvement nobody made is worse than no arrow.

Both numbers are each curve's **true** peak, deliberately not the trimmed range the axis is drawn to. They disagreed once: the pill read 35.61 → 77.13 above a curve that visibly reached 180. The axis is the thing allowed to trim; a curve leaving the top of the plot is how it says so.

03 The chip row that holds it

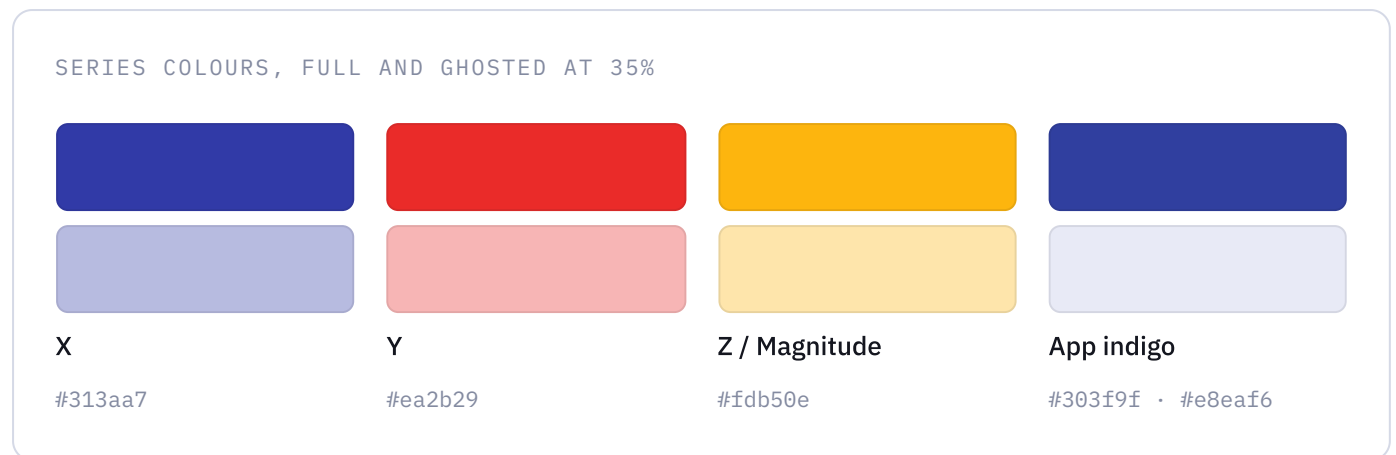
Sits directly above the chart, flat on the card — a note about what is drawn, not a control sitting on top of it.



The swatch is **grey on purpose**, not a series colour: it stands for *all* the ghosted curves, and a graph can show three at once, so there is no one colour to borrow. What it actually means is “pale”.

04 The palette it borrows from

The overlay does not introduce colour. It reuses the plot's own three series and the app's indigo.



The amber row is the one to look at hardest — see question 2.

05 Related treatments

These shipped alongside and share the vocabulary, so they are worth seeing together.

WHERE	TREATMENT
Start-pose ghost canvas	Where the machine's cycle starts, under the live mechanism at 22% opacity — same shapes, same colours, plainly the same object earlier. 40% on hover; pressing it returns to the start. When a drag takes the linkage somewhere its original start no longer exists it becomes a dotted outline with hollow pins at 50% , in informational indigo rather than an alarm colour — nothing is broken, letting go just moves the start. A pill tag rides the pointer to say so.
Drag trace canvas	While you hold a joint, its full path over the cycle is drawn — literally the app's existing <i>Traced Paths</i> curve, same ink, same weight. It was a custom dashed partial arc until this week.
Transport row bottom bar	A hollow dashed ring marks where the cycle starts on the scrub rail. A <code>134° from start</code> chip appears when you are parked away from it; pressing it goes back, and a caret opens “Move the start here”. A <code>Start moved here</code> chip with an inline Undo records the change.
Refusal strip edit panel	When the panel cannot be typed into: a full-bleed 54px strip in <code>#e8eaf6</code> on <code>#303f9f</code> directly under the title, with the way out written as an underlined word inside the sentence rather than a button beside it. Same height in all five states.
Corner card top strip	Undo and Redo live there in Edit. Export Data slides in from the right (220ms) when you enter an analysis mode, animating width and margin together so the card grows <i>with</i> it and the two buttons beside it never jump.

06 Where I would like your eye

Ranked by how unsure I am. Everything here is a decision I made alone and would happily lose.

1 The green / amber pill makes a value judgement

A smaller peak is usually what someone wants — but not always, and colouring it good or bad may be overstepping for a teaching tool. Is a neutral delta better?

2 35% alpha is very faint on the amber series

Against white, #fdb50e ghosted is much weaker than the same treatment on indigo. Should ghosting be a fixed grey, a fixed alpha, or tuned per colour?

3 Two encodings change at once on release

Dashed becomes solid at the same moment the gesture ends. That may be one change too many — is ghosting alone enough to carry “before”, with the live curve solid throughout?

4 The chip row competes for its slot

An amber “No solution at N of M positions” banner can occupy the same strip above the chart. They never show at once today, but they are two different shapes in one place.

5 The peak numbers carry no unit

The unit lives on the axis only, so peak 35.61 → 21.04 is ambiguous read on its own.

6 Width is not fixed

The panel is 250px in Edit and 400px in analysis, and on a phone it is a bottom sheet. The chip row can wrap at the narrow width.

7 Indigo is doing a lot of jobs

The ghost, the refusal strip, the start chip, the seat and the X series are all indigo. One vocabulary, or an overload?

07 Seeing it live

Everything above is running at staging--pmksnew.netlify.app (<https://staging--pmksnew.netlify.app>).

1. Open any template, then switch to **Kinematic Analysis**.
2. Press Play, then pause mid-cycle.
3. Click a joint to choose what is graphed.
4. Drag a *different* joint. Click selects; drag tunes.